HACK MATRIX HACKATHON

TITLE PAGE

- **PROJECT TITLE : Rangmanch -** Local Artisan Marketplace

- **TEAM NAME : AC-DC**

- **TEAM MEMBERS :**

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PROBLEM STATEMENT

- **Problem Description:**
Rural artisans produce high-quality, authentic handmade goods but lack the digital literacy and infrastructure to access global or even urban markets.

- **Affected Parties:**
Low-income rural artisans, NGOs working in craft preservation, and conscious consumers looking for authentic products.

- **Importance:**
Without a digital sales channel, traditional crafts are dying out as artisans move to daily-wage labor for financial survival.

EXISTING SOLUTIONS AND GAPS

- **Existing Solutions:**
Etsy, Amazon Karigar, Facebook Marketplace.

- **Limitations:**
High commission fees, complex onboarding processes for non-tech-savvy users, and a lack of direct communication (WhatsApp) for negotiation.

- **The Gap:**
RangManch provides a "Zero-Barrier" entry point with simplified listing, regional filtering, and direct-to-WhatsApp buyer-seller connection

PROPOSED SOLUTION

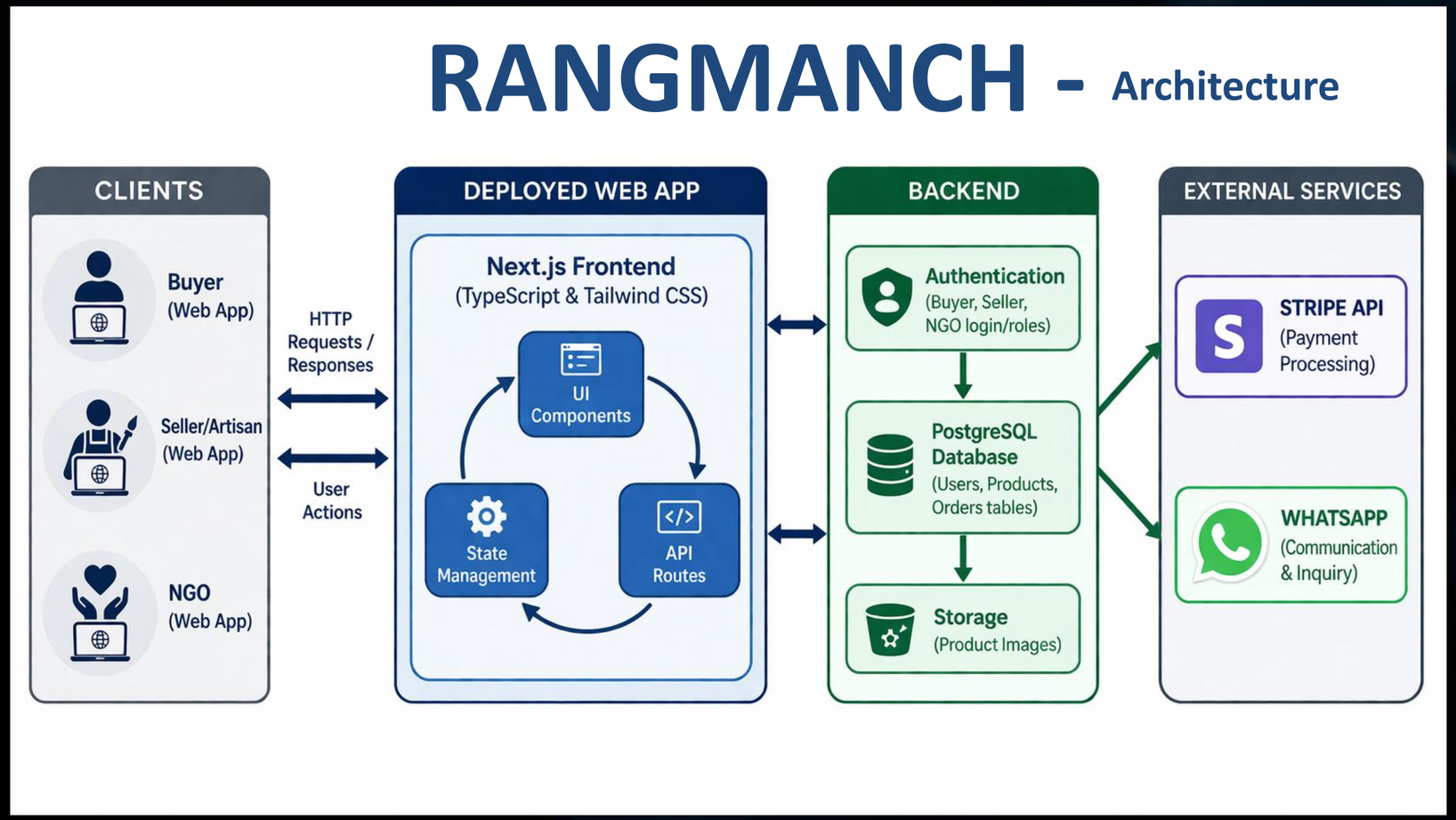
- **Explanation:** A specialized marketplace for three user roles:
  - **Sellers** (Artisans) to list products
  - **Buyers** to purchase
  - **NGOs** to facilitate and verify rural sellers.

- **Key Features:** * Role-based Login (Buyer/Seller/NGO).
- One-click Product Listing with photo upload.
- Search by Category or Region.
- Dual-Action Purchase: Stripe for secure payments and WhatsApp for direct negotiation.

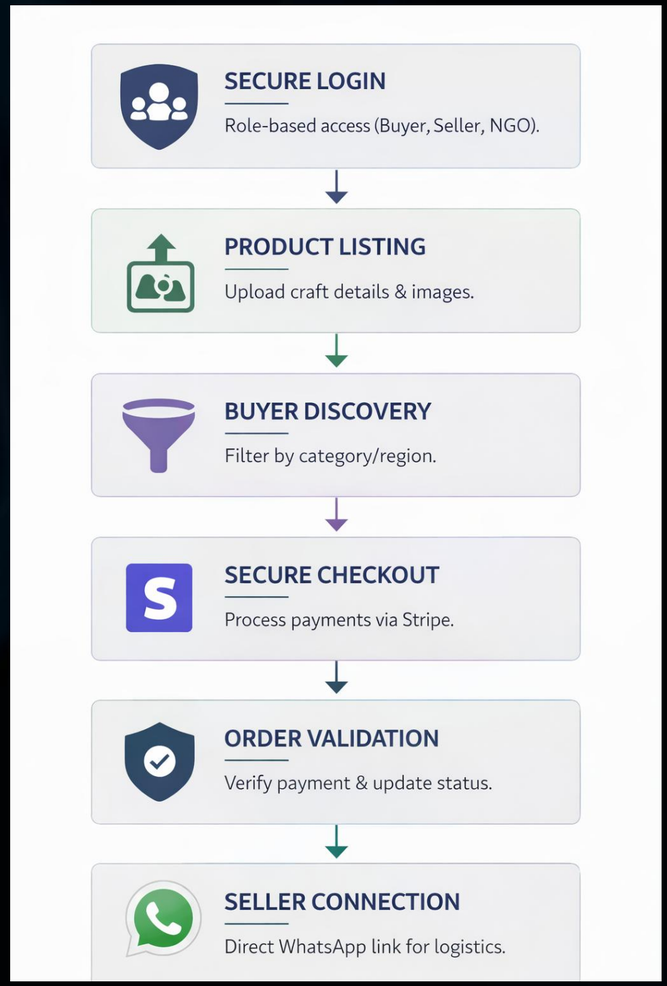
- **Problem Solving**

It bridges the digital divide by making the "storefront" as easy to manage as a messaging app.

SYSTEM ARCHITECTURE

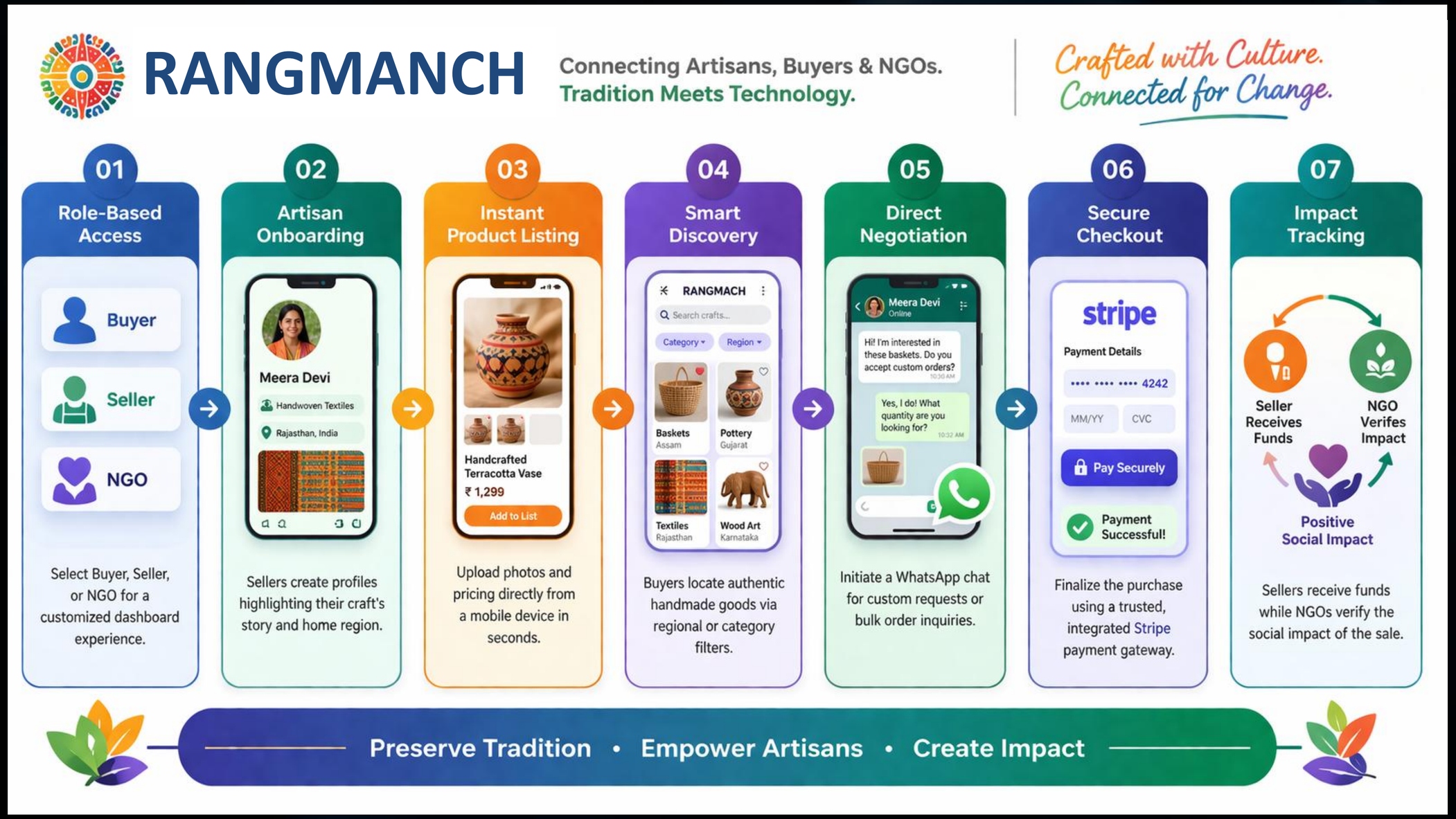


Architecture Diagram



Data Flow

FLOWCHART / WORKING PROCESS



TECH STACK

- **Frontend: Next.js (TypeScript & Tailwind CSS)**
  - **Server-Side Rendering (SSR):** Critical for SEO, ensuring artisan products appear in search engine results to drive organic traffic.
  - **Built-in Image Optimization:** Automatically serves compressed, high-quality images of handmade crafts, essential for fast loading on rural/mobile networks.

- **Backend: Node.js with Express.js**
  - **Asynchronous Architecture:** Handles thousands of concurrent buyer-seller interactions without blocking the server, essential for high-traffic marketplace events.
  - **Middleware Ecosystem:** Utilizes professional security layers like **Helmet.js** for header protection and **CORS** for secure cross-origin communication.

- **Database: PostgreSQL (via Supabase)**
  - **Relational Integrity:** Uses a robust relational model to strictly manage complex links between Artisans, their Product listings, and NGO verifications.
  - **Scalability:** A production-grade database capable of handling thousands of concurrent transactions as the platform grows.

- **Tools & APIs**
  - **Stripe API:** Provides a globally trusted, PCI-compliant gateway for secure buyer payments and automated seller payouts.
  - **Vercel:** Offers a high-performance hosting environment with an integrated CI/CD pipeline for seamless project updates.

IMPLEMENTATION PLAN

Development Plan

Setup
Init Next.js, Tailwind CSS & PostgreSQL schema.

Auth
Role-based login for Buyers, Sellers & NGOs.

Inventory
Seller dashboard for product/photo uploads.

Discovery
Buyer search with regional & category filters.

Commerce
Stripe API & "Chat on WhatsApp" button.

Verification
NGO profile validation & impact tracking.

Launch
Deploy to Vercel for live hosting.

Core Modules & Features

User Management
Secure role-based profiles (Buyer/Seller/NGO).

Listing Engine
Simplified product uploads with image optimization.

Search Hub
Geo-regional & category-based craft discovery.

Communication
Direct WhatsApp bridge for negotiation.

Payment Gateway
PCI-compliant Stripe transactions.

Impact Tracking
NGO-verified artisan badges.

Timeline

0–8h: Foundation
Init Next.js & Auth Setup.

8–16h: Seller MVP
Listing Forms & Cloudinary Uploads.

16–24h: Discovery
Buyer Search & Filters.

24–32h: API Integration
Stripe & WhatsApp Connect.

32–36h: Verification
NGO Dashboard & Badges.

36–40h: Deployment
Vercel Launch & Final QA.

EXPECTED OUTCOME / IMPACT

- **Expected Results:**
  - **Fully Functional MVP:** A production-ready platform featuring role-based dashboards and a <60s "Instant Listing" workflow.
  - **Secure Ecosystem:** 100% successful payment processing via **Stripe** and verified seller-buyer communication via **WhatsApp**.

- **Benefits to Users:**
  - **Sellers (Artisans):** Direct market access, cutting out exploitative middlemen to increase profit margins by **30-40%**.
  - **Buyers:** Guaranteed authenticity through **NGO-verified** listings and unique, traceable handmade products.
  - **NGOs:** Centralized data for tracking rural economic growth and identifying artisan needs.

- **Real-World Impact:**
  - **Economic Sustainability:** Transforming seasonal craftwork into a consistent, scalable digital business.
  - **Preservation:** Revitalizing dying art forms by making them financially viable for the younger generation.

FUTURE SCOPE AND CONCLUSION

- **Future Improvements:**
  - **AI-Driven Studio:** Automated background removal and lighting enhancement for professional-grade product photos.
  - **Vernacular Voice UI:** Voice-to-Text onboarding in local dialects to assist non-literate or tech-shy artisans.
  - **AR Product Preview:** Augmented Reality integration allowing buyers to "visualize" crafts in their own homes before purchase.

- **Scalability of the Project:**
  - **Logistics Partnerships:** Integration with hyper-local delivery services for automated rural-to-urban shipping.
  - **B2B Channel:** Expanding into bulk corporate gifting and international exports for certified handmade goods.

SUMMARY

Rangmanch acts as a digital bridge for rural artisans, transforming every local smartphone into a global storefront. By enabling direct commerce through secure payments and WhatsApp, we empower craftsmen to preserve their heritage while achieving financial independence. Rangmanch turns ancient tradition into sustainable modern opportunity, ensuring our cultural heartbeat thrives powerfully—one authentic, precious handmade treasure at a single time.